Week 1: Practice exercises

Complete this exercise in a separate notebook. Write the program first, then test it using the suggested inputs.

Exercise 1: Even or Odd Checker

Task. Ask the user for a whole number and print whether it is even or odd.

REQUIREMENTS

- Read a number from the user.
- Convert the input to an integer.
- Use an if/else decision.
- Print exactly one clear result: Even or Odd.

TEST YOUR PROGRAM

Try this input	Expected result
8	Even
13	Odd
0	Even

Hint: An even number has a remainder of 0 when divided by 2. The modulo operator (%) gives you the remainder.

Exercise 2: Favorite Color Match

Task. Create a list of colors such as ["red", "blue", "green"]. Ask the user for a color and report whether it appears in the list.

REQUIREMENTS

- Create the list before asking for input.
- Use the in operator to test membership.
- Print a message for both a match and a non-match.

TEST YOUR PROGRAM

Try this input	Expected result
red	The color is in the list.
purple	The color is not in the list.

Hint: The expression color in colors evaluates to True or False.

Exercise 3: Positive, Negative or Zero

Task. Ask the user for a number. Print Positive when it is greater than 0, Negative when it is less than 0, and Zero when it is exactly 0.

REQUIREMENTS

- Read a numeric value from the user.
- Use if, elif and else.
- Handle all three possible cases.

TEST YOUR PROGRAM

Try this input	Expected result
7	Positive
-2	Negative
0	Zero

Hint: Test the greater-than condition first, then the less-than condition. If neither is true, the value must be zero.

Exercise 4: Age Group

Task. Ask the user for their age. If the age is less than 18, print Minor; otherwise print Adult.

REQUIREMENTS

- Convert the age to an integer.
- Use the condition age < 18.
- Use else for the adult case.

TEST YOUR PROGRAM

Try this input	Expected result
12	Minor
17	Minor
18	Adult
25	Adult

Hint: The boundary value matters: age 18 belongs to the else branch.

Exercise 5: Pass or Fail

Task. Ask the user for an exam mark. If the mark is 50 or higher, print Pass; otherwise print Fail.

REQUIREMENTS

- Read the mark as a number.
- Use the comparison >= 50.
- Print Pass or Fail.

TEST YOUR PROGRAM

Try this input	Expected result
49	Fail
50	Pass
76	Pass

Hint: Use >= rather than > so that a mark of exactly 50 passes.

Exercise 6: Smallest Number Check

Task. Create a list containing five numbers. Ask the user for another number and check whether it is smaller than every number in the list.

REQUIREMENTS

- Create a list with exactly five numeric values.
- Read the user number.
- Compare it with the smallest list value.
- Print a clear True/False-style message.

TEST YOUR PROGRAM

Try this input	Expected result
List: [12, 7, 20, 4, 9], input: 3	Smaller than all numbers
List: [12, 7, 20, 4, 9], input: 4	Not smaller than all numbers
List: [12, 7, 20, 4, 9], input: 10	Not smaller than all numbers

Hint: If a number is smaller than every value in a list, it must be smaller than min(numbers).

Exercise 7: Weekend or Weekday

Task. Ask the user to enter a day of the week. Print Weekend for Saturday or Sunday; otherwise print Weekday.

REQUIREMENTS

- Read the day as text.
- Check Saturday and Sunday together in the condition.
- Print Weekend or Weekday.

TEST YOUR PROGRAM

Try this input	Expected result
Monday	Weekday
Saturday	Weekend
Sunday	Weekend

Hint: You can check whether the day is in a small collection: day in ("Saturday", "Sunday").